

CampusEats - Assignment 3

Integrate an External SOAP Partner

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1 Context

1.1 Partner: PayFlex Payment Gateway

Operation Integrated: *charge* - authorizes and captures a card payment for a CampusEats order at checkout.

The checkout/payment edge is the one place in CampusEats where **SOAP** is justified while the rest of the system (menu browsing, cart, order status, delivery tracking) stays **REST**.

1. **Strong, machine-checkable contract:** The WSDL fixes the exact request/response/fault shapes so both CampusEats and PayFlex can generate client/server stubs and catch mismatches at design time rather than at runtime.
2. **Message-level security:** The merchant credentials and card token travel inside a signed SOAP header (WS-Security style) that stays protected even if the message is logged or relayed through an intermediary, which plain transport-level TLS on a REST call does not guarantee end-to-end.
3. **Transactional, exactly-once operation:** A charge operation with a formal fault contract (soap:Fault for declined/unavailable) matches how payment gateways are conventionally integrated in industry.

The rest of CampusEats is simple resource CRUD (GET/POST a menu item, an order, a cart) with no such contractual or security weight, so lightweight REST/JSON is the better fit there.

2 HTTP Binding

The request below is the concrete HTTP POST that carries the charge SOAP envelope, matching the `soap:address` in `partner.wsdl` and the `soapAction` in the binding.

```
POST /soap/v1/payments HTTP/1.1 Host: api.payflex.example.com Content-Type:
text/xml; charset="utf-8" SOAPAction: "http://payflex.example.com/payments/v1/charge"
Content-Length: 683
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:pf="http://payflex.example.com/payments/v1"> <soapenv:Header>
<pf:Credentials> <pf:apiKey>sklive51NcampuseATS9f2 </pf : apiKey >< pf :
merchantId > CAMPUSEATS - MERCH - 0042 </pf : merchantId >< pf :
timestamp > 2026 - 08 - 26T10 : 15 : 30Z </pf : timestamp >< pf : nonce >
a1b2c3d4 - e5f6 - 4a7b - 8c9d - 0e1f2a3b4c5d </pf : nonce >< /pf : Credentials ><
/soapenv : Header >< soapenv : Body >< pf : ChargeRequest >< pf : merchantId >
CAMPUSEATS - MERCH - 0042 </pf : merchantId >< pf : orderId >
ORD - 20260826 - 00931 </pf : orderId >< pf : amount > 458.00 </pf : amount ><
pf : currency > INR </pf : currency >< pf : cardToken > tok9f8e7d6c5b4a </pf :
cardToken >< /pf : ChargeRequest >< /soapenv : Body >< /soapenv : Envelope >
```

Request Details:

- **Method:** POST
- **Host:** api.payflex.example.com
- **Content-Type:** text/xml; charset="utf-8"
- **SOAPAction:** "http://payflex.example.com/payments/v1/charge"
- **Endpoint:** /soap/v1/payments (from service/port in partner.wsdl)

3 Discovery (Modern Form)

CampusEats does not run a live UDDI server. Instead the partner WSDL is obtained in one of two standard modern ways, and the fact of the integration is recorded once in CampusEats' own internal API catalogue:

- Direct URL:** PayFlex publishes its contract at a stable, versioned developer-portal URL (<https://developer.payflex.example.com/wsdl/payments/v1/partner.wsdl>) that CampusEats' integration team downloads once during partner onboarding and pins in the repo (this file).
- Catalogue/Registry Lookup:** CampusEats also records the partner as one entry in its internal service catalogue (e.g. Backstage, an internal API registry, or even a shared `partners.json`), so any team member can look up "who do we call for payments" without re-discovering the URL.

3.1 Registry/Catalogue Entry

Field	Value
businessEntity	PayFlex Inc.
businessKey	payflex-inc
serviceName	PayFlex Payment Gateway - charge
endpoint	payflex-payments-v1
wsdlTModel	https://api.payflex.example.com/soap/v1/payments
bindingStyle	document/literal (SOAP 1.1 over HTTP)
category	payment-gateway
registeredBy	CampusEats Integration Team
lastVerified	2026-08-26

4 Fault Mapping

PayFlex's fault vocabulary is internal to PayFlex and must **never leak** to CampusEats students/clients. The SOAP adapter catches the soap:Fault from PayFlex, logs the partner's raw faultCode/declineReason server-side only, and re-raises it as the plain error CampusEats' own Assignment 2 placeOrder contract already promised.

PayFlex fault (partner vocabulary)	CampusEats placeOrder error (our contract)
card_declined	402 PAYMENT_DECLINED – “Your payment could not be completed. Please try another card.”
gateway_unavailable / timeout	503 SERVICE_UNAVAILABLE – “Payment service is temporarily unavailable, please retry.”
invalid_card_token	402 PAYMENT_DECLINED – same generic student-facing message as card_declined

Example: The soap-fault.xml sample (Task 3) carries faultCode=card_declined with a partner-specific declineReason. The adapter maps this to a single stable **PAYMENT_DECLINED** error in the CampusEats API response - the client never sees PayFlex's internal codes or wording.